1. The Core Research Problem: Deeper Elaboration

The Crisis: In India, we have high-tech hospitals but low health literacy. A patient gets a report saying *"Right-sided pleural effusion,"* but they tell the doctor, *"Doctor, chhati me pani bhar gaya hai right side pe."*
The AI Failure: Current LLMs (GPT-4o, Llama 3) treat "Hinglish" as a generic language. They lack the **Medical Latent Space** knowledge for Indian vernacular.
Your Contribution: You are researching **Fact-Grounded Vernacular Reasoning**. You aren't just making a chatbot; you are researching how a machine can "prove" a Hinglish statement is true by looking at a X-ray.

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2. Detailed Methodology & Challenges

A. Data Construction: The "Synthetic-to-Real" Pipeline

- **Detailed Task:** You take 10,000 MIMIC-CXR radiology reports. You use an LLM to generate **"Intent Pairs."**
    - *Input:* "Cardiomegaly with pulmonary edema."
    - *Generated Hinglish Query:* "Dil bada ho gaya hai aur lungs me sujan lag rahi hai, kya ye serious hai?"
- **Challenge: "Robotic Translation."** Synthetic Hinglish often sounds too formal or like Google Translate, which doesn't reflect how real patients in Mumbai/India speak.
- **How to Overcome:** Use **"Few-Shot Contextual Prompting."** Feed the LLM examples from Indian medical forums (like Practo/1mg) so it learns the *social dialect* of Indian patients (e.g., using "tension," "report," "sujan").

B. Cross-Modal Alignment (The Encoding Engine)

- **Detailed Task:** You use **BioMedCLIP** for images and **mBERT/LaBSE** for text. You must align them in a **Shared Vector Space**.
- **Challenge: "Semantic Drift."** The vector for the English word "Pneumonia" and the Hinglish phrase "Lungs me infection" might end up in different parts of the database, causing the RAG to retrieve the wrong image.
- **How to Overcome:** Implement **"Contrastive Fine-Tuning."** You "pull" the Hinglish vector and the corresponding BioMedCLIP image vector closer together using a Triplet Loss function. This forces the model to learn that these two different formats represent the same medical reality.

C. The Multimodal RAG Orchestration

- **Detailed Task:** When a query comes in, you retrieve the Image + the original English report (Ground Truth).
- **Challenge: "Context Overflow."** Feeding a high-resolution image and a long text report into a model like **LLaVA-Med** can exceed the "Context Window" or make the model ignore the text in favor of the image.
- **How to Overcome:** Use **"Visual Prompt Tuning."** Instead of the whole image, you extract "Medical Visual Tokens" (the most important parts of the scan) to keep the prompt concise and focused.

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3. Answering Critical Research Questions (The Defense)

During your viva or paper review, these questions **will** arise. Here are the researched answers:

Q1: Why use RAG? Why not just fine-tune LLaVA-Med on Hinglish data?

- **Answer:** Fine-tuning is "Static." It only knows what it was trained on. RAG is "Dynamic." In a real hospital, data changes every day. RAG allows the model to look at the **specific patient's** newest scan and history, which it has never seen before, ensuring the answer is personalized, not just a general guess.

Q2: How do you evaluate if the "Hinglish" generated is actually correct?

- **Answer:** Standard metrics like BLEU or ROUGE are insufficient for medicine. We propose a **"Clinical Fact-Check Metric."** We will extract the medical entities from the Hinglish output (using a NER model) and compare them against the entities in the English Ground Truth. If the English says "Effusion" and the Hinglish says "Sujan" (Swelling), the score drops.

Q3: How will the system handle "Language Dominance" (e.g., 90% English, 10% Hindi)?

- **Answer:** This is the core of our **Linguistic Resilience** research. We will test the retriever's performance at varying levels of "Code-Mixing Index" (CMI). Our goal is to prove that by using Multimodal embeddings (Vision + Text), the system remains stable even when the text grammar is broken.

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4. Implementation Challenges & Overcoming Them

Challenge	Impact	How to Overcome
Data Access (PhysioNet)	You can't download the data without approval.	**Act Now:** All 3 members should finish the **CITI Data Ethics Training** (it takes 3 hours) and apply for MIMIC-CXR access immediately.
GPU Compute Power	LLaVA-Med is 7B parameters. It might lag on a standard laptop.	**Quantization:** Use **4-bit/8-bit Quantization (bitsandbytes)** to run the model on a single 16GB GPU (like Kaggle or Google Colab T4).
Alignment Errors	Retrieving a lung scan when the patient asks about their heart.	**Re-ranking:** After the first retrieval, use a "Cross-Encoder" to double-check the relevance of the top 3 images before showing them to the LLM.

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5. What makes this "The Best" for your group?

1. **Complexity:** It shows you can handle **Image Processing**, **Advanced NLP**, and **System Architecture (RAG)**.
2. **Uniqueness:** There are 0 papers currently that specifically study **Multimodal RAG for Hinglish Radiology**.
3. **Job Ready:** By the end, Member 1 is a Computer Vision expert, Member 2 is a GenAI/LLM expert, and Member 3 is a Backend/MLOps expert.

This is a comprehensive, step-by-step roadmap for your research project: **Vision-Grounded Multimodal RAG for Code-Switched Hinglish Clinical Decision Support**.

Phase 1: Research Foundation & Data Access

Goal: Secure the clinical data and establish the research questions.

- **Step 1.1: Ethical Approval & Data Sourcing:** You must use the **MIMIC-CXR-JPG v2.0.0** dataset (approx. 377,000 X-ray images with reports).
    - *Task:* All 3 members must complete the **CITI Data Ethics Training** (free) to get access via PhysioNet.
- **Step 1.2: Define Research Questions (RQs):**
    - **RQ1:** Can a cross-modal retriever accurately align informal Hinglish semantics with specific radiographic visual tokens?
    - **RQ2:** To what extent does multimodal retrieval-augmentation reduce factual hallucinations in Hinglish diagnostic summaries compared to baseline LLMs?

- **RQ3:** How does the "Code-Mixing Index" (ratio of Hindi to English) affect the precision of medical entity retrieval?

Phase 2: The Hinglish-Medical-Grounding (HMG) Dataset Construction

Goal: Build the first-of-its-kind dataset that links Hinglish queries to X-rays.

- **Step 2.1: Semantic Parsing:** Extract 10,000 "Findings" and "Impressions" from the MIMIC English reports.
- **Step 2.2: Synthetic-to-Real Synthesis (NLP Lead):**
  - Use **GPT-4o** or **Llama-3.1-405B** to transform English findings into **Hinglish Intent Pairs**.
  - *Prompting Strategy:* Use "Few-shot prompting" with real-world examples from the **HEALTH-PARIKSHA** dataset to ensure the code-switching sounds like a Mumbai patient (e.g., using words like *takleef*, *sujan*, *report clear hai*).
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- **Step 2.3: Metadata Mapping:** Create a JSON structure where each Hinglish query points to a specific **Image ID** and the **Segmented visual area** (e.g., lower left lobe).

Phase 3: Cross-Modal Alignment (The "Encoding Engine")

Goal: Make the "Hinglish word" and the "X-ray pixels" talk the same mathematical language.

- **Step 3.1: Visual Encoding (Vision Lead):**
  - Use **BioMedCLIP**.[1][2] Pass the X-rays through its Vision Transformer (ViT) to get **768-dimensional visual vectors**.
- **Step 3.2: Linguistic Encoding (NLP Lead):**
  - Use a multilingual model like **LaBSE** or **mBERT**. These models are already good at keeping *"Chest pain"* and *"Chhati me dard"* close in vector space.
- **Step 3.3: Contrastive Fine-Tuning (Research Core):**
  - This is your "Research Gap" solution. You will fine-tune the text encoder so that the vector for the Hinglish query *"Left side me congestion"* is mathematically "pulled" toward the BioMedCLIP visual vector of a lung with opacity.
  - *Math:* Use **InfoNCE Loss** to maximize the similarity between matching pairs and minimize it for non-matching ones.

Phase 4: Multimodal RAG Pipeline Orchestration

Goal: Build the retrieval logic and the reasoning engine.

- **Step 4.1: Vector Indexing (System Lead):**
  - Insert your 10,000 X-ray vectors and English report vectors into **ChromaDB**.
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- **Step 4.2: Retrieval Logic:**
  - When a user types a Hinglish query, turn it into a vector, search ChromaDB, and retrieve the **Top-3 most similar X-rays** and their corresponding **English Ground Truths**.
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- **Step 4.3: Grounded Generation (LLaVA-Med):**
  - Create a "Multimodal Prompt": [Hinglish Query] + [Retrieved X-ray Image] + [Retrieved English Text Context].
  - **LLaVA-Med** processes this.[3] The "English Context" acts as an anchor to prevent the model from hallucinating in Hinglish.

Phase 5: Evaluation & Stress Testing

Goal: Prove your system is better and safer.

- **Step 5.1: Comparative Baselines:** Compare your results against **GPT-4o (Zero-shot)** and **Standard English RAG**.
- **Step 5.2: Clinical Fact-Checking Metric:**
  - Don't just use BLEU/ROUGE. Use **RadGraph F1-Score**. This measures if the "Medical Entities" (like *Effusion* or *Cardiomegaly*) mentioned in your Hinglish output match the English Ground Truth.
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- **Step 5.3: Hallucination Audit:** Manually check 50 cases where the AI disagreed with the report. Analyze *why* it failed (e.g., was the Hinglish too slangy?).

Phase 6: Deployment (The Final Web App)

Goal: A demo-ready application for your Semester 3/Placements.

- **Tech:** Use **Streamlit** for the frontend and **FastAPI** for the backend.
- **Feature: Evidence Highlighting.** When the model says *"Left side pe opacity hai,"* use the attention maps from LLaVA-Med to draw a **bounding box** on the X-ray, showing the user exactly what the AI is looking at.

Challenges & How to Overcome Them

Challenge	Impact	Mitigation Strategy
GPU Limitations	Models like LLaVA-Med are heavy.	Use **4-bit Quantization** (using the bitsandbytes library) to run it on a single 16GB GPU.
Translation Bias	Synthetic Hinglish can sound robotic.	Use the **IHQID dataset** to extract real-world patient query templates and fill them with medical entities from MIMIC.
Alignment Drift	The model retrieves the wrong organ.	Implement a **Re-ranker**. After retrieving the top 10 images, use a smaller classifier to filter only those that match the requested organ (e.g., Chest).

Summary of Research Value

By following these steps, you are not just building a "chatbot." You are researching **Cross-Modal Factuality**. You are proving that medical AI can be **inclusive** (Hinglish) without being **dangerous** (Hallucinations). This is the exact profile **Big Tech AI Labs** are hiring for in 2026.

The "Wow Factor" of your research project lies in its **uniqueness and impact**. In a world full of generic chatbots, you are building a **"Localized Expert"** that solves a high-stakes problem no global tech giant has fully cracked yet: **Multilingual Medical Trust.**

Here is the breakdown of the "Wow Factor" and how to pitch it to different audiences.

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1. The Three Layers of the "Wow Factor"

- **The "Vernacular White Space":** Global AI is trained on English medical textbooks. You are building the **first-of-its-kind dataset (HMG)** that maps the actual way Indians speak (Hinglish) to high-tech radiology.
- **The Hallucination "Kill-Switch":** Standard AI "hallucinates" medical facts. Your project uses **Multimodal RAG** as a safety valve. The AI isn't just "talking"; it is **grounded**—it can't make a claim without pointing to a specific patch of pixels in an X-ray.
- **The "Local Hero" Impact:** You are addressing the Indian **Radiologist Shortage** (1 per 100,000 people). This isn't just a paper; it's a tool for rural healthcare workers who speak Hinglish and need to interpret complex scans instantly.

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2. Explaining to a Technical Domain (Recruiters, Engineers, Professors)

Focus on architecture, data novelty, and mathematical alignment.

> "Our research tackles **hallucination mitigation in Medical LVLMs** by implementing a **Vision-Grounded Multimodal RAG** pipeline. The core innovation is our **Cross-Modal Alignment** strategy: we developed a **Synthetic-to-Real pipeline** to create the HMG dataset, mapping code-switched Hinglish clinical intent to radiographic visual features. Technically, we utilize **Contrastive Learning** (via BioMedCLIP) to align the latent spaces of informal Hinglish semantics with formal visual tokens. We don't just generate text; we use **RadGraph-F1 metrics** to ensure our agentic reasoning is grounded in retrieved visual evidence, achieving a high degree of **clinical factuality** that standalone models lack."

Key Technical Buzzwords to use:

- *Cross-Modal Grounding*
- *Latent Space Alignment*
- *Code-Mixing Index (CMI) Resilience*
- *Hallucination Audit*

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3. Explaining to a Non-Technical Domain (Parents, Friends, HR Managers)

Focus on the "Why," social impact, and trust.

> "Imagine a patient in an Indian village who gets a complex X-ray report written in difficult medical English. They might tell the doctor, *'Chhati me bohot tension hai, report me kya dikh raha hai?'* Current AI often makes mistakes or gives dangerous advice because it doesn't understand this mix of Hindi and English. **My project**

builds a 'Safe Translator' for doctors. It looks at the actual X-ray, reads the English report, and explains it in simple Hinglish. It acts like an expert who 'sees' the problem and 'speaks' the patient's language, making sure the AI never lies or makes things up. We are basically giving every village clinic an expert assistant who speaks like a local."

Key Analogies to use:

- **The "Fact-Checker":** Compare it to a lawyer who isn't allowed to speak in court unless they are pointing at a specific piece of evidence.
- **The "Cultural Bridge":** Compare it to a bridge that connects high-tech city hospitals with rural patients through the common language of Hinglish.

Summary Table: The "Elevator Pitch"

Audience	The "Hook"
Technical	"We are solving the cross-modal grounding gap for Hinglish clinical diagnostics using agentic RAG."
Non-Technical	"We are building a life-saving AI that understands Hinglish and can't lie about medical reports."
Recruiter	"I built a multimodal system that handles the three hottest trends of 2026: **LLMs, Vision, and RAG.**"

When researchers and medical professionals review your proposal for **Vision-Grounded Multimodal RAG for Hinglish Clinical Intelligence**, they will look for "cracks" in the logic, specifically regarding medical safety and linguistic complexity.

Here are the top 5 critical questions they will ask, along with the sophisticated answers your group should provide.

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1. The "Safety" Question

Question: *"RAG systems are only as good as what they retrieve. What happens if the system retrieves the wrong X-ray (a False Positive) and then 'forces' the AI to explain it to the patient? Won't this create a 'Grounded Hallucination' that is even more dangerous because it looks credible?"*

Answer:
"This is a primary concern in clinical AI. To mitigate this, we implement a **Triple-Verification Layer**:

1. **Confidence Thresholding:** If the retrieval similarity score (Cosine Similarity) is below a specific threshold (e.g., 0.85), the system is programmed to 'Abstain' rather than guess.
2. **Cross-Check Logic:** We perform a text-to-text check between the user's query and the retrieved report *before* the image is shown to the model.
3. **Human-in-the-loop:** The system is designed as a 'Decision Support' tool, not a diagnostic one. It clearly displays the retrieved source text alongside the answer, allowing a medical professional to verify the evidence immediately."

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2. The "Linguistic" Question

Question: *"Hinglish is not a standard language; it varies by region, education level, and context. How do you ensure your 'Synthetic Hinglish' dataset doesn't suffer from 'Linguistic Drift' where it fails on real-world, messy human dialects?"*

Answer:
"We address this by moving beyond simple rule-based translation. Our dataset construction uses **Persona-Based Prompting** informed by the **Code-Mixing Index (CMI)**. We pull syntactic templates from authentic Indian health forums (like Practo/1mg) and populate them with medical entities from MIMIC. We don't just 'translate' English to Hinglish; we 're-simulate' the query based on real-world patient behavior patterns. Furthermore, we use **LaBSE (Language-Agnostic BERT Sentence Embedding)**, which is specifically designed to be resilient to code-switching and 'noisy' grammar."

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3. The "BioMedCLIP" Question

Question: *"BioMedCLIP's latent space was aligned using formal English academic papers (PubMed). By forcing it to align with informal Hinglish, aren't you risking 'Catastrophic Forgetting'—where the model loses its high-level medical precision to gain linguistic flexibility?"*

Answer:
"To prevent this, we utilize **Parameter-Efficient Fine-Tuning (PEFT)**, specifically **LoRA (Low-Rank Adaptation)**. We keep the core weights of the BioMedCLIP Vision Encoder and the PubMedBERT backbone frozen. We only train a lightweight **Projection Layer** (the 'adapter') that maps Hinglish embeddings into the existing medical latent space. This ensures the model retains its 15-million-image medical 'wisdom' while simply learning a new 'vernacular interface' to access that wisdom."

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4. The "Clinical Validity" Question

Question: *"Standard NLP metrics like BLEU or ROUGE only measure if the sentence 'sounds right.' How will you prove that the Hinglish output is medically accurate? A sentence can be 90% grammatically correct but 100% medically wrong."*

Answer:
"We agree that traditional NLP metrics are insufficient for healthcare. Our research uses **RadGraph F1-Score** and **CheXbench**.

- **RadGraph:** We extract 'Clinical Entities' (e.g., 'Lung Opacity') and 'Relations' (e.g., 'located in Left Lobe') from the Hinglish output and compare them to the English Ground Truth.
- **Safety Audit:** We specifically measure the 'Clinical Error Rate'—the percentage of times a generated Hinglish summary contradicts the primary diagnosis in the retrieved English report."

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5. The "Scalability" Question

Question: *"X-rays in rural India are often low-quality, taken with portable machines, or have different lighting conditions compared to the 'clean' MIMIC-CXR dataset. How will your system handle this 'Domain Shift'?"*

Answer:
"While this research focuses on the linguistic-visual bridge, we incorporate **Image Augmentation** (Gaussian noise, motion blur, and contrast shifts) during our retrieval testing to simulate real-world 'noisy' scans. In the future, this system can be integrated with **Unsupervised Domain Adaptation (UDA)** to further 'normalize' low-quality images before they enter the RAG pipeline. Our current scope establishes the multimodal framework, which can then be hardened for varying image qualities."

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6. The "Modality Imbalance" Question

Question: *"In your Multimodal RAG, what happens if the modalities conflict? For example, if the retrieved English report says 'Normal Heart' but the user's Hinglish query claims 'Dil me sujan' (Swelling in heart). Which one does the AI trust, and how do you prevent the model from getting confused?"*

Answer:
"This is the challenge of **Multimodal Conflict Resolution**. We implement a **Weighted Fusion Mechanism**.

1. Since the English report is the 'Clinical Ground Truth' (validated by a radiologist), the system is instructed to give it higher 'Semantic Weight.'
2. If a conflict is detected, the AI is programmed to use a **'Clarification Loop.'** Instead of giving a final diagnosis, the output will state: *'Report ke mutabik heart normal hai, par aapne sujan ka zikr kiya hai. Kya aap confirm kar sakte hain ki ye purani report hai ya nayi takleef?'*
3. This ensures the AI acts as a **verifier** rather than a blind generator."

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7. The "Medical Negation" Question

Question: *"In Hinglish, negation is tricky. A patient might say 'Chest me pain nahi hai' versus 'Chest me pain hai.' Standard encoders often ignore small words like 'nahi.' How do you ensure your RAG doesn't retrieve 'Pneumonia' reports for a patient who explicitly says they DON'T have symptoms?"*

Answer:
"This is a classic 'Negation Scope' problem in NLP. To solve this, we use **Negation-Aware Embeddings**:

1. Instead of just using a global sentence vector, we use a **Clinical NER (Named Entity Recognition)** layer (like Med-SpaCy) to specifically tag 'Negated Entities.'
2. During retrieval, we use **Hard Negative Mining**. We specifically train the model to distinguish between *'X-ray me pneumonia dikh raha hai'* and *'X-ray me pneumonia nahi hai.'*
3. This prevents the 'Bag-of-Words' fallacy where the model only looks for the keyword 'Pneumonia' and ignores the context."

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8. The "Inference Latency" Question

Question: *"A Multimodal RAG pipeline (Encoding + Vector Search + LLaVA-Med Reasoning) is computationally expensive. In a busy Mumbai OPD (Outpatient Department), a doctor can't wait 30 seconds for an AI response. How do you optimize for real-time speed?"*

Answer:
"Latency is a critical bottleneck for clinical adoption. We use three optimization strategies:

1. **Vector Quantization (Product Quantization):** We compress the BioMedCLIP vectors in ChromaDB so that the search takes milliseconds, even across 100,000 records.
2. **KV-Caching & Speculative Decoding:** For the LLaVA-Med generation, we use KV-caching to avoid re-calculating visual tokens for every word generated.
3. **Asynchronous Pre-fetching:** While the user is still typing their Hinglish query, the system begins pre-encoding the uploaded image, so the retrieval starts the moment they hit 'Enter'."

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9. The "Expert-in-the-Loop" Question

Question: *"As Data Science students and not Medical Doctors, how can you claim your 'Synthetic Hinglish' is medically valid? Who is the 'Gold Standard' for your evaluation?"*

Answer:
"We acknowledge our role as technical researchers, not clinicians. Therefore, we use **Proxy Gold Standards**:

1. We use the **MIMIC-CXR 'CheXpert' Labels** as our ground truth. These are labels generated by a high-accuracy automated labeler that has been validated by a panel of board-certified radiologists.
2. We evaluate our system's 'Reasoning' by checking if it recovers the **same CheXpert labels** from the Hinglish query that the original English report had.
3. For the linguistic part, we use **Hinglish-fluent evaluators** (including our group) to check for 'Semantic Naturalness' using the Code-Mixing Index (CMI)."

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10. The "Bias & Fairness" Question

Question: *"Medical reports for different demographics (Age, Gender) use different terminology. Is there a risk that your Hinglish RAG performs better for young urban males than for elderly rural females due to the way they code-switch?"*

Answer:
"This is a question of **Socio-Linguistic Fairness**.

1. We conduct an **Ablation Study across CMI levels**. We test the model on 'Heavy Hindi' Hinglish and 'Heavy English' Hinglish to see if accuracy drops.
2. Our use of **BioMedCLIP** helps mitigate this because the *visual data* (the X-ray) remains objective, regardless of how the patient speaks.
3. By 'anchoring' the generation in the image, we ensure that even if the Hinglish query is grammatically 'non-standard,' the AI still has the visual evidence to stay on track."